

COMP 2121 – Quiz 5

Time: 20 min.

Possible marks: 5

Name:

Student #:

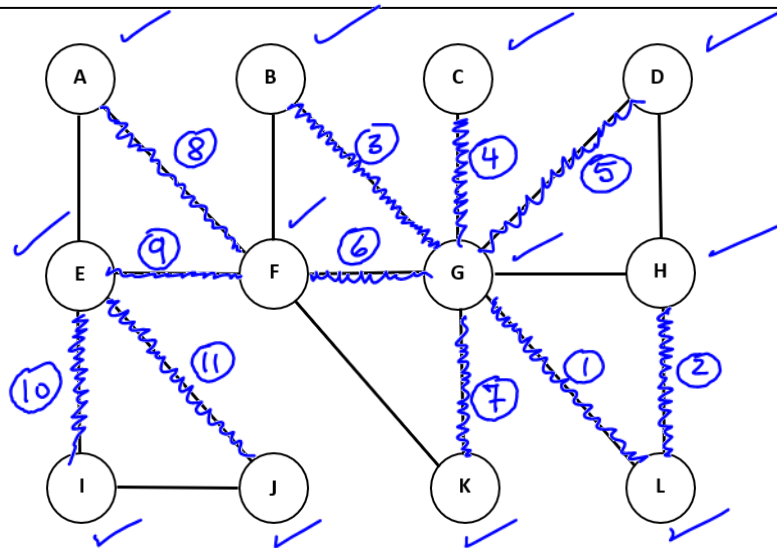
1. (2 marks) Run BFS algorithm **starting at vertex L** and:

- Show its spanning tree. Give a label to each edge according to the order it is added to the spanning tree (i.e. edge that is added first label 1, edge that is added second label 2, etc.). Also,
- Write the order in which vertices were added to the queue Q.

Breadth – First Search Algorithm.

Insert vertex v_x at the rear of (initially empty) queue Q and initialize T as the tree made up of this one vertex v_x . Visit v_x .

```
While (Q is not empty) {  
    Delete the vertex v from the front of the Q  
    for each neighbor w of v //in the increasing order  
        if w is unvisited  
        {  
            visit(w);  
            insert w at the rear of the Q;  
            add edge vw to tree T  
        }  
}
```



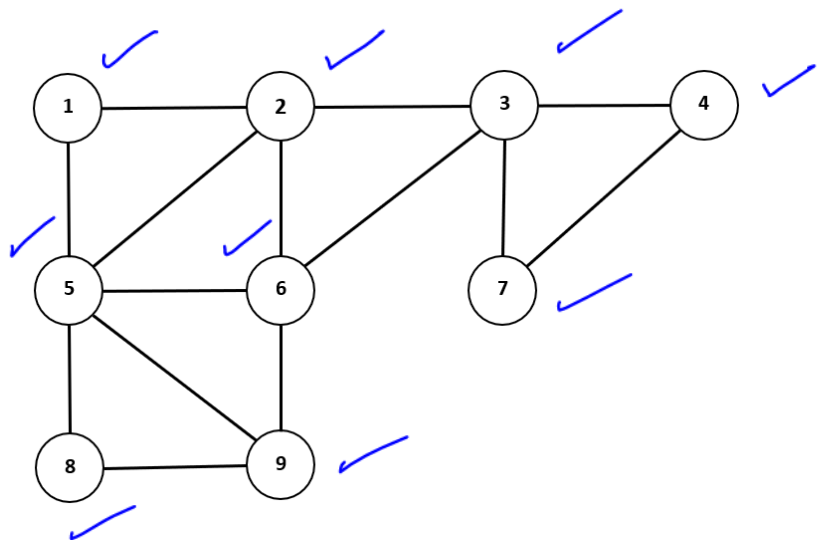
Q: ~~L~~, ~~G~~, ~~H~~, ~~B~~, ~~C~~, ~~D~~, ~~F~~, ~~K~~, ~~A~~, ~~E~~, ~~I~~, ~~S~~

2. (3 marks) Do modified DFS on the following graph, **starting at vertex 2**:
For full marks, you must include both the correct Return Stack and the list of vertices printed on the screen.

```
dfs(vertex v)
{
    visit(v);
    for each neighbor w of v //in the increasing order
        if w is unvisited
            {
                print v
                dfs(w);
                print w
            }
}
```

Show the following:

1) RETURN STACK



2) PRINTED ON THE SCREEN:

[illegible][illegible]